

University of Madras
Department of Computer Science
Data Analytics – TEST - I

Section A (4 X 5 = 20 marks) Answer any 4

1. Explain the 5Vs of data analysis with suitable examples.
2. Elaborate on exploratory data analysis with example for each.
3. Illustrate SVM.
4. What is predictive analysis? Give example and its evaluation metrics.
5. What is Regression used for? Explain with example.
6. Describe neural networks usage for prediction.
7. With python codes explain data preprocessing methods with examples.
8. Generate decision tree for the following data

Age	Competition	Type	profit
Old	Yes	Software	Down
Old	No	Software	Down
Mid	Yes	Software	Down
Old	No	Hardware	Down
Mid	Yes	Hardware	Down
Mid	No	Hardware	Up
Mid	No	Software	Up
New	Yes	Software	Up
New	No	Hardware	Up
New	No	Software	Up

University of Madras
Department of Computer Science
Data Analytics
End Semester

Max Time : 3Hrs.

Max Marks : 60

Section A (10 X 1 = 10 marks)

Answer any 10

1. Define data analytics. Name the types of data for analysis.
2. Expand OLAP, SNA, RTAP, SAAS
3. List any four big data technologies used for data analytics
4. What is sensemaking ?
5. Name any four data sources with example.
6. Write the names of various analytic classes.
7. Differentiate between classification and clustering .
8. Define Association rule. Give example
9. When to use ensemble model? Give example.
10. Enumerate the performance metrics of a classification method.
11. What is the use of entropy and gini index ?
12. Justify the need of collective inference in data analytics, with an example.

Section B (5 X 10 = 50 marks)

Answer all

- 13 a. Describe the methods used for data preprocessing and its treatment.

OR

- 13 b. (i) Explain any five sampling methods for data analysis (5) (ii) List the types of cloud services and give example (5)

- 14 a. Give an account on predictive analytics using regression, with example.

OR

- 14 b. Write short notes on (i) neural networks (5) (ii) support vector machine (5)

- 15 a. How patterns are discovered using Apriori algorithm? Write the algorithm and explain with an example.

OR

- 15 b. Elaborate on (i) segmentation as data analytics method (5) (ii) survival analysis (5)

- 16 a. Write about (i) Web Analytics (5) (ii) Model design and documentation (5)

OR

- 16 b. With python code, explain any two data analytics methods and its inference component.

- 17 a. Discuss on (i) Business intelligence tools (5) (ii) Hadoop for Data Analytics (5)

OR

- 17 b. Write a python program with any five graphical visualization techniques suitable for Social Network Analysis and explain them.